Notes about the Final Exam for 159.102 (Computer Science and programming)

The final exam will be a **closed-book, in-person, supervised digital exam**.

- **Date:** Tuesday, 27 October 2026
- **Duration:** 3 hours
- Completed on a **computer provided at the venue** (no internet or external software access)
- **Content covered:** Week 5 – week 11

  Main focus of the final exam: MASSEY machine, Classes, Pointers, Recursion, Data Structures, Linked lists, and a small amount of graphics.

Please check your **exam venue and time** via your student portal and email. If you have not received this information, contact **Assessment Services (exams@massey.ac.nz)**.

If you are unable to attend your exam, contact **Assessment Services** as soon as possible to discuss possible arrangements.

Key reminders

- Bring **photo ID** (student ID, passport, etc.) and **pens**
- **No** laptop required (devices provided)
- **Draft paper** will be provided and collected
- **Calculators are allowed** (scientific or graphical), but they may not be needed

All exams are managed by Assessment Services, not the teaching team.

More details about exams at Massey:
https://stream.massey.ac.nz/course/view.php?id=79449

Preparing for the Final Exam

There are six (6) questions. Some questions contain multiple parts (a), (b), (c), etc.

The final exam is marked out of 50 and contributes 40% towards your final course result.

Tips for each question:

Question 1 (6 marks)

Study the **Week 5** notes, workshops, and programming practice exercises. The **MASSEY machine instruction set** will be provided in the exam.

Question 2 (7 marks)

Study the **Week 6** and **Week 10 (Stacks)** notes, workshops, and programming practice exercises. You should know:

- how to use pointers
- how to use call by reference
- how to use stacks

Question 3 (6 marks)

Study the **Week 8** notes, workshops, and programming practice exercises.
You should know:

- how to use recursion
- how to compare the **Linear Search** and **Binary Search** algorithms by counting the number of comparisons
- how to implement backtracking without creating an infinite loop

Note: You will **not** be required to design a new recursive program.

Question 4 (6 marks)

Study the **Week 9** notes, workshops, and programming practice exercises.
Note: You will **not** be required to design any new images.

Question 5 (15 marks)

Study the **Week 7** and **Week 10** notes, workshops, and programming practice exercises.
Ensure that you are familiar with:

- designing new classes
- using vectors

You will be required to **write a complete program** for this question.

Question 6 (10 marks)

Study the **Week 7** and **Week 11** notes, workshops, and programming practice exercises.
You will be required to identify **five (5) errors** in a program. The program will include **classes and linked lists**.

Time management advice

Do not waste time: First, read through the entire exam and allocate an appropriate amount of time to each question. If your allocated time for a question runs out, move on to the next one and return to it later if time permits.

There is little benefit in completing Question 1 perfectly if you do not have enough time to attempt the remaining questions.

How to calculate your final result in 159.102

Assignment 1 = 15%
Assignment 2 = 20%
Assignment 3 = 15%
Quiz 1 (Test) = 10%
Quiz 2 (Final exam) = 40%

Total for the course = 100%

You will pass the course if your **overall course mark is 50% or higher**.
NOTE: Your final course result is **not** calculated on Stream. It is calculated separately.